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## 安全工程专业培养方案

### Curriculum for Undergraduate of Safety Engineering Major

#### 一、专业介绍

安全工程专业始建于1987年，1993年和2004年分别开始培养硕士和博士研究生，2011年获国家首批安全科学与工程一级学科博士学位授予权，2019年获批博士后科研流动站，2021年通过国家工程教育认证，2022年获批国家一流本科专业建设点。本专业拥有湖北省工业安全工程技术研究中心、工业安全数智化企校联合创新中心等平台，建有通风防尘、职业危害控制、防火防爆、矿山安全、安全人机、安全检测、安全仿真等专业实验室。本专业毕业生超过40%继续深造，大部分毕业生就业于中钢集团、中建集团、中交集团、中铁集团等大型国有企事业单位或其他知名企业。

#### I. Major Introduction

Safety Engineering major was founded in 1987, beginning as the Mine Ventilation and Safety major. It started training master's and doctoral students in 1993 and 2004, respectively. In 2011, it was awarded the first batch of doctoral degree in Safety Science and Engineering. In 2019, it was approved as a postdoctoral research station. In 2021, it passed the national engineering education certification. In 2022, it was approved as the national first-class undergraduate professional construction site. It has platforms such as the Hubei Industrial Safety Engineering Technology Research Center and the Industrial Safety Digitalization Enterprise School Joint Innovation Center, as well as specialized laboratories for ventilation and dust prevention, occupational hazard control, fire and explosion prevention, mining safety, safety human-machine interaction, safety testing, and safety simulation. More than 40% of graduates in this major continue to pursue graduate studies. The majority of graduates are working in large state-owned enterprises and institutions such as China Steel Group, China Construction Group, China Communications Construction Corporation, and China Railway Group, or other well-known enterprises.

#### 二、培养目标

面向国家发展战略、社会经济发展需求和行业发展趋势，培养德智体美劳全面发展，拥有人文素养、科学精神、实践能力、创新创业精神和国际视野，掌握安全科学、安全工程及技术的基础理论和专业知识，具备危险有害因素识别、风险管控、应急管理的能力，能够在金属矿山、冶金、建筑等行业及政府部门、科研院所从事安全科技研究、安全设计、安全监察与管理、应急技术与管理、安全咨询与评价、安全教育与培训等岗位的学术型人才。

本专业毕业生毕业五年左右预期达到以下目标：

- (1) 具有良好的品德修养和职业道德，积极服务于国家和社会，成为社会主义事业建设者和接班人；
- (2) 具备注册安全工程师的素质，具有应用专业知识分析问题、设计技术方案、解决复杂安全工程问题的能力；
- (3) 具备良好的沟通、合作和领导能力，具有一定的国际视野，能在团队中担任骨干或负责人；
- (4) 具有终身学习的能力，掌握行业前沿动态和发展趋势，适应职业发展的要求。

## II. Training objectives

This major educated high quality applied talents can satisfy the need of socialist market economy construction, develop comprehensively in morality, intelligence and physique etc., master the basic theory and professional knowledge of safety science, safety engineering and technology, acquire fundamental training as a safety engineer, be capable of safety system design, safety technical consultation, safety education, safety management and supervision in metal mines and metallurgical manufacturing, possess innovation and entrepreneurship and international view, and adapt to the transformation and upgrading of industrial safety.

Graduates of this major are supposed to achieve the following aims after five years:

1. Possessing good moral cultivation, professional ethics and healthy physical and mental, being active service to the country and society, and to become a builder and successor of the socialist cause;
2. Mastering the fundamental quality and ability of certified safety engineer, being capable of safety technology development, safety engineering design, safety evaluation and enterprise safety management in metal mines and metallurgical manufacturing, and to become a business backbone or technical leader;
3. Having well communication, coordination and leadership ability, with good international view, and to become a core leader in a multi-disciplinary and multi-cultural safety engineering project;
4. Paying attention to the cutting-edge dynamics of safety science and technology and the development trend of emergency management continuously, adapting to the requirements of intelligent and digital industrial safety, having the sense of lifelong learning, and to be occupational competitive in the field of safety engineering.

## 三、专业特色

本专业秉持工程教育专业认证理念，开展国家级一流本科专业建设，以金属矿山与冶金制造行业安全为主线，以解决复杂工程问题为目标，融合智能仿真技术，强化“五育并举”和“三全育人”，培养学生的工程实践与创新创业能力。

## III. Professional characteristics

Safety Engineering Major adheres to the concept of engineering education professional certification, conducts the construction of the National First - class Undergraduate Major. Taking the safety of the metal mining and metallurgical manufacturing industries as the main line, aiming to solve complex engineering problems, integrating intelligent simulation technology, strengthening the “simultaneous promotion of the five - aspect education” and “education through all - around efforts”, and cultivating students’ engineering practice, innovation, and entrepreneurship abilities.

## 四、毕业要求

- (1) 工程知识：能够将数学、自然科学、安全工程基础和专业知用于解决金属矿山、冶金、建筑等行业的复杂安全工程问题。

(2) 问题分析：能够应用数学、自然科学和工程科学的基本原理，识别、表达、并通过文献研究分析金属矿山、冶金、建筑等行业生产过程中的复杂安全工程问题，以获得有效结论。

(3) 设计/开发解决方案：能够针对机械安全、通风除尘和建筑防火等复杂工程进行规划和方案设计，并能够在规划、设计环节体现创新意识，考虑健康、安全与环境、全生命周期成本与净零碳要求、法律与伦理、社会与文化等因素。

(4) 研究：能够基于安全工程学科的基本原理并采用科学的方法对复杂安全工程问题进行研究，具有设计实验、分析与解释数据，并融合数据智能处理技术得到合理有效结论的能力。

(5) 现代工具的应用：能够应用计算机程序对工程进行辅助设计，应用安全模拟仿真实论和技术对安全工程及相关领域的复杂工程问题进行分析、预测和模拟，并能够理解其局限性。

(6) 工程与社会可持续发展：能够基于安全工程相关知识进行合理分析，评价生产、建设工程实践和复杂安全工程问题解决方案对社会、健康、安全、法律以及经济和社会可持续发展的影响，并理解应承担的责任。

(7) 工程伦理和职业规范：树立和践行社会主义核心价值观，具有推动社会主义事业建设的责任感和使命感，有工程报国、为民造福的意识，具有人文社会科学素养、社会责任感，能够在安全工程实践中理解并遵守工程职业道德和规范，履行责任。

(8) 个人与团队：能够在多样化、多学科背景下的团队中承担团队成员以及负责人的角色。

(9) 沟通：能够就复杂安全工程问题与业界同行及社会公众进行有效沟通和交流，包括撰写报告和设计文稿、陈述发言、清晰表达或回应指令。具备一定的国际视野，能够在跨文化背景下进行沟通和交流，并理解、尊重语言和文化差异。

(10) 项目管理：理解并掌握工业生产中的安全管理原理和相关的经济决策方法，并能在多学科环境中应用。

(11) 终身学习：能够持续关注国内外工业安全及相关领域的前沿动态和发展趋势，具有自主学习和终身学习的意识，有不断学习和适应发展的能力。

#### IV. Requirements

1. Engineering knowledge: With the ability of using mathematics, natural sciences, basic and professional knowledge in engineering to solve complex engineering problems in the field of safety engineering.

2. Problem analysis: Identifying, expressing, and analyzing the complex engineering problems in safety engineering area with the literature research methods to obtain effective conclusions, basing on the application of basic principles of mathematics, natural sciences and engineering science.

3. Design/develop solutions: Providing solutions solve the complex problems in the field of safety engineering. Designing systems, units (components) or processes meet specific needs. With the ability of embodying innovation awareness as well as the factors of health, safety, environment, life-cycle cost, net-zero carbon requirements, law, ethics, society and culture, etc, in the designing process.

4. Experimental design and information processing: Researching complex engineering problems in the field of safety engineering by scientific methods based on scientific principles, including experiments designing, data analysis and interpretation, to obtain reasonable and effective conclusions through comprehensive information analysis.

5. Using modern tools: With the ability of developing, selecting and using appropriate technologies, resources, modern engineering tools, information technology tools to solve complex engineering issues in the field

of safety engineering, including the prediction and simulation of complex engineering problems and understanding its limitation.

6. Engineering and social sustainability: With the ability of making reasonable analysis based on the relevant engineering background, to evaluate the impact of the engineering practice and complex engineering solutions in the field of safety engineering on society, health, safety, legal, culture and economic and social sustainability as well as understanding the corresponding responsibility.

7. Engineering ethics and professional norms: With high sense of humanities and social science literacy, social responsibility, having the awareness of serving the country through engineering and bringing benefits to the people, understanding and observing engineering ethics and norms in safety engineering and related engineering practice.

8. Individual and team: Be able to play a role as a team member or a team manager in diversity and multi-disciplinary environment. 9. Communication: Be able to communicate effectively with industry peers and the public on complex engineering issues in the field of safety engineering, including writing reports, designing documents, presenting statements, clearly expressing or responding to directives. Communicating and exchanging ideas within the cross-cultural background with international perspective, and understand and respect linguistic and cultural differences.

10. Project management: With the ability of understanding and mastering the management principles and economic decision-making methods in the field of safety engineering, and be able to apply them in multi-disciplinary environment.

11. Lifelong time learning: With the sense of independent learning and lifelong time learning awareness, as well as the continuous learning and development adaptive ability.

附：培养目标实现矩阵（表1）和毕业要求与观测指标点（表2）

表1 培养目标实现矩阵

|         | 培养目标 1 | 培养目标 2 | 培养目标 3 | 培养目标 4 |
|---------|--------|--------|--------|--------|
| 毕业要求 1  |        | √      |        |        |
| 毕业要求 2  |        | √      |        |        |
| 毕业要求 3  |        | √      |        |        |
| 毕业要求 4  |        | √      |        | √      |
| 毕业要求 5  |        | √      |        | √      |
| 毕业要求 6  | √      |        |        |        |
| 毕业要求 7  | √      |        |        |        |
| 毕业要求 8  |        |        | √      |        |
| 毕业要求 9  |        |        | √      |        |
| 毕业要求 10 |        |        | √      |        |
| 毕业要求 11 |        |        |        | √      |

表2 毕业要求与观测指标点

|                                                                                                                            |                                                               |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 1. 工程知识：能够将数学、自然科学、安全工程基础和专业用于解决金属矿山、冶金、建筑等行业的复杂安全工程问题。                                                                    | 1-1：掌握所需的数学知识，能针对具体的安全工程问题建立数学模型并求解。                          |
|                                                                                                                            | 1-2：能将物理、化学等自然科学知识用于安全工程问题及事故致因的阐释。                           |
|                                                                                                                            | 1-3：掌握能够将安全工程基础知识和技术用于推演、分析复杂工程问题。                            |
|                                                                                                                            | 1-4：综合应用所学工程知识，用于金属矿山、冶金、建筑等行业的复杂安全工程问题解决方案的比较和综合。            |
| 2. 问题分析：能够应用数学、自然科学和工程科学的基本原理，识别、表达、并通过文献研究分析金属矿山、冶金、建筑等行业生产过程中的复杂安全工程问题，以获得有效结论。                                          | 2-1：能够应用数学和自然科学基本方法，将生产过程中的复杂工程问题转化为数学和力学问题进行表述、分析。           |
|                                                                                                                            | 2-2：能够利用专业知识识别和判断复杂安全工程问题的关键环节。                               |
|                                                                                                                            | 2-3：在实践教学过程中，面对复杂安全工程问题能提出多种解决方案。                             |
|                                                                                                                            | 2-4：具备通过文献研究分析和判断复杂安全工程问题并获得有效结论的能力。                          |
| 3. 设计/开发解决方案：能够针对机械安全、通风除尘和建筑防火等复杂工程进行规划和方案设计，并能够在规划、设计环节体现创新意识，考虑健康、安全与环境、全生命周期成本与净零碳要求、法律与伦理、社会与文化社会、健康、经济、法律、文化以及环境等因素。 | 3-1：掌握机械安全、通风除尘、防火防爆工程、职业危害控制工程的设计流程和技术方法，了解影响设计目标和技术方案的各种因素。 |
|                                                                                                                            | 3-2：能够完成机械安全、通风除尘和建筑防火等工程单元设计，在设计中体现创新意识。                     |
|                                                                                                                            | 3-3：具有安全工程实践学习的经历，在规划和设计中能够独立提出方案、完成报告，同时具有进行创新性规划和设计的能力。     |
|                                                                                                                            | 3-4：了解并掌握企业生产和施工建设中的社会、健康、安全、法律、文化以及环境等多方面要求，在安全工程规划和设计中得到满足。 |
| 4. 研究：能够基于安全工程学科的基本原理并采用科学的方法对复杂安全工程问题进行研究，具有设计实验、分析与解释数据，并融合数据智能处理技术得到合理有效结论的能力。                                          | 4-1：熟练掌握主要安全测试和实验仪器的知识、理论、原理及使用技能。                            |
|                                                                                                                            | 4-2：能够基于安全科学原理与方法，针对复杂安全工程问题，设计实验方案，并开展实验研究，获取实验数据。           |
|                                                                                                                            | 4-3：应用数理方法、数据智能处理技术等分析及解读数据，通过信息综合得到合理有效的结论。                  |
| 5. 现代工具的应用：能够应用计算机程序对工程进行辅助设计，应用安全模拟仿真理论和技术对安全工程及相关领域的复杂工程问题进行分析、预测和模拟，并能够理解其局限性。                                          | 5-1：掌握计算机辅助设计相关软件，并能熟练应用于相关工程设计。                              |
|                                                                                                                            | 5-2：掌握火灾和烟气模拟仿真技术，能对安全工程领域相关问题进行模拟、预测和优化，并能够理解其局限性。           |
|                                                                                                                            | 5-3：了解现代数据分析方法，能使用相关数据处理软件分析和优化复杂工程问题，并能够理解其局限性。              |

|                                                                                                        |                                                                                     |
|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 6. 工程与社会可持续发展：能够基于安全工程相关知识进行合理分析、评价生产、建设工程实践和复杂安全工程问题解决方案对社会、健康、安全、法律以及经济和社会可持续发展的影响以及文化的影响，并理解应承担的责任。 | 6-1：具备从事安全评价的能力，熟练掌握安全生产相关法律法规、风险辨识与评估、风险控制的方法与理论。                                  |
|                                                                                                        | 6-2：能运用相关知识分析和评价生产、建设工程实践和复杂工程问题解决方案对社会、健康、安全以及经济的影响，并理解应承担的责任。                     |
|                                                                                                        | 6-3：具备评价生产过程和建设项目对环境、社会可持续发展影响的能力。                                                  |
| 7. 工程伦理和职业规范：树立和践行社会主义核心价值观，具有推动社会主义事业建设的责任感和使命感，具有人文社会科学素养、社会责任感，能够在安全工程实践中理解并遵守工程职业道德和规范，履行责任。       | 7-1：具有人文社会科学素养、思辨能力和科学精神，身心健康。                                                      |
|                                                                                                        | 7-2：树立和践行社会主义核心价值观，理解个人与社会的关系，了解国情，有工程报国、为民造福的意识，掌握基本的军事知识和技能，具有推动社会主义事业建设的责任感和使命感。 |
|                                                                                                        | 7-3：理解诚实、公正、守信的职业道德，具有较高的安全职业素养，能够在安全工程及相关工程实践中遵守工程职业规范，履行责任。                       |
| 8. 个人与团队：能够在多学科背景下的团队中承担团队成员以及负责人的角色。                                                                  | 8-1：具备团队意识和合作精神。                                                                    |
|                                                                                                        | 8-2：能够根据自身作为团队成员或团队负责人的不同角色，做好本职工作，发挥个体在团队中作用。                                      |
| 9. 沟通：能够就复杂安全工程问题与业界同行及社会公众进行有效沟通和交流，包括撰写报告和设计文稿、陈述发言、清晰表达或回应指令。并具备一定的国际视野，能够在跨文化背景下进行沟通和交流。           | 9-1：具备就安全工程领域复杂工程问题与业界同行及社会公众进行有效沟通和交流的能力。                                          |
|                                                                                                        | 9-2：能够进行安全工程研究分析报告、设计文稿和公务文等文件的撰写。                                                  |
|                                                                                                        | 9-3：能熟练应用至少一门外语，具备一定的国际视野，能够在跨文化背景下进行沟通和交流，并理解、尊重语言和文化差异。                           |
| 10. 项目管理：理解并掌握工业生产中的安全管理原理和相关的经济决策方法，并能在多学科环境中应用。                                                      | 10-1：掌握项目管理的方法和原理，具备建设工程及相关工程安全监理和决策的能力。                                            |
|                                                                                                        | 10-2：了解工程项目和工业产品生产的全周期、全流程中与安全相关的经济决策问题。                                            |
|                                                                                                        | 10-3：能将安全管理原理与经济决策方法在金属矿山、冶金、建筑等行业的安全工程问题中应用。                                       |
| 11. 终身学习：能够持续关注国内外工业安全及相关领域的前沿动态和发展趋势，具有自主学习和终身学习的意识，有不断学习和适应发展的能力。                                    | 11-1：具有自主学习和终身学习的意识。                                                                |
|                                                                                                        | 11-2：通过不断自主学习，具备适应工业智能化和数字化发展要求的能力。                                                 |

### 五、专业主干课程

安全科学导论、安全法规、安全管理、安全人机工程学、安全系统工程、安全检测与智能监测、防火防爆安全技术、锅炉压力容器安全技术、职业危害与防治。

### V. Core courses

Introduction of Safety Science, Safety Laws/Regulations, Safety Management, Safety Ergonomics, Safety System Engineering, Safety Detection and Intelligent Monitoring, Fire Prevention and Explosion Protection Technologies, Boiler and Pressure Vessel Safety Technology, Occupational Hazards and Prevention.

### 六、基本学制：四年

### VI. Recommended length of the program: 4 years

### 七、授予学位：工学学士

### VII. Degree: Bachelor of Engineering

学生修满所规定的最低毕业学分，符合武汉科技大学授予学士学位规定，授予工学学士学位。

### 八、毕业学分要求：160学分

| 课程类型     |    | 学分要求 | 课程类型     |    | 学分要求 |
|----------|----|------|----------|----|------|
| 1、公共课程平台 |    | 44   | 3、专业课程模块 |    | 44.5 |
| 公共基础课程   |    | 28   | 专业必修课程   |    | 23   |
| 通识教育课程   | 必修 | 10   | 专业选修课程   |    | 21.5 |
|          | 选修 | 6    | 4、实践教学模块 |    | 23.5 |
| 2、学科基础平台 |    | 42   | 专业实践课程   | 必修 | 23.5 |
| 专业学科基础课程 | 必修 | 34.5 |          |    |      |
|          | 选修 | 7.5  | 5、素质拓展模块 |    | 6    |

\*通识教育选修课6学分包括：理工类、医学类、人文社科类、经济管理类中选择2学分（学生在本专业所属类别外的3个类中选择2个学分）；思想政治及新时代素质教育类选择2学分（“四史”至少选修1个学分）；美育教育类选择2学分（美学和艺术史论类、艺术鉴赏和评论类课程至少选修1个学分）。

### VIII. Credits required for graduation : 160 credits

| Type of courses                |                  | Academic credits | Type of courses                    |                  | Academic credits |
|--------------------------------|------------------|------------------|------------------------------------|------------------|------------------|
| 1.Common Courses               |                  | 44               | 3. Specialized Courses             |                  | 44.5             |
| Common Basic Courses           |                  | 28               | Required Courses                   |                  | 23               |
| General Education Courses      | Required Courses | 10               | Elective Courses                   |                  | 21.5             |
|                                | Elective Courses | 6                | 4.Practicum and Internship Courses |                  | 23.5             |
| 2.General Disciplinary Courses |                  | 42               | Disciplinary Practical Courses     | Required Courses | 23.5             |
| Disciplinary Basic Courses     | Required Courses | 34.5             |                                    |                  |                  |
|                                | Elective Courses | 7.5              | 5.Quality Development Courses      |                  | 6                |

## 九、学分比例

### IX. Ratio of Credits

#### 1、必修选修学分比例

The proportion of compulsory elective credits

| 类别 | 学分  | 占总学分比例  |
|----|-----|---------|
| 必修 | 125 | 78.125% |
| 选修 | 35  | 21.875% |

#### 2、实践教学环节学分比例

The Proportion of credits in practice teaching

| 类别     |        | 学分   | 占总学分比例 |
|--------|--------|------|--------|
| 实践教学环节 | 实验教学学分 | 13   | 26.6%  |
|        | 实践教学模块 | 23.5 |        |
|        | 素质拓展模块 | 6    |        |

## 十、辅修

修读本专业辅修课程达到26学分且主修专业达到毕业要求者，颁发辅修专业证书。修读本专业辅修课程达到40学分（含实践教学），并达到辅修学位授予条件的，颁发辅修学位证书。

### X. Minor

The minor certificate will be issued to those who have completed 26 credits of minor courses of their own major and their major has met the graduation requirements. The minor degree certificate will be issued to those who have completed 40 credits of minor courses (including practical teaching) and meet the conditions for awarding minor degree.

## 十一、毕业要求实现矩阵

## XI. Graduation Realization Matrix

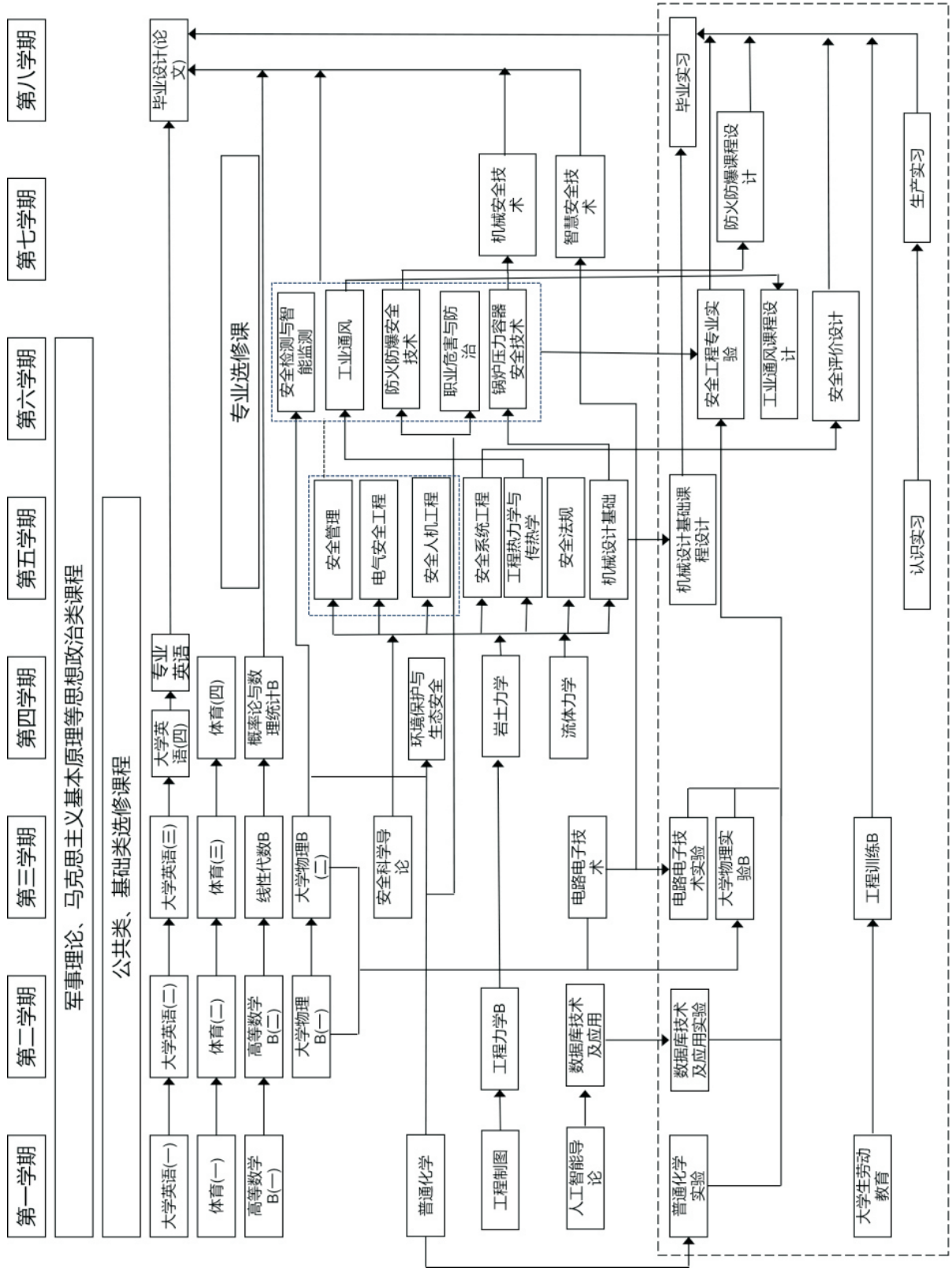
| 课程名称   |                      | 安全工程专业毕业要求指标点 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|--------|----------------------|---------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
|        |                      | 1             | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 5 | 5 | 5 | 6 | 6 | 6 | 7 | 7 | 7 | 8 | 8 | 9 | 9 | 9 | 1 | 1 | 1 | 1 |   |   |
|        |                      | -             | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | 0 | 0 | 0 | 1 | 1 |   |
|        |                      | 1             | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 1 | 2 | 3 | 1 | 2 | 3 | 1 | 2 | 3 | 1 | 2 | 1 | 2 | 3 | - | 1 | 2 | 3 | 1 | 2 |
| 公共课程平台 | 思想道德修养与法律基础          |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | M | M |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 中国近现代史纲要             |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | M |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 马克思主义基本原理            |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | M |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 毛泽东思想与中国特色社会主义理论体系概论 |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | M |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 国家安全教育               |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | M |   |   |   | H |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 人工智能导论               |               |   |   |   |   |   |   |   |   |   |   |   |   |   | H | H |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | M |   |
|        | 大学英语                 |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | H |   |   |   |   |   | L |   |
|        | 体育                   |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | H |   |   | M |   |   |   |   |   |   |   |   |   |   |
|        | 大学生心理健康教育            |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | H |   |   |   |   |   |   |   |   |   |   | H |   |   |
|        | 职业生涯规划与就业创业指导        |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | M |   |   |   |   |   |   |   | L |   |   |
|        | 军事理论                 |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | H |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 大学生劳动教育              |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | L |   |   | H |   |   |   |   |   |   |   |   |   |   |
|        | 形式与政策                |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | M |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 创业学基础                |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | H |   | H |   |   |   |   |   |   | M |   |
| 学科基础平台 | 工程制图                 |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   | H |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 机械设计基础               |               |   |   |   |   |   |   |   | M | L |   |   |   |   |   |   |   | M |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 线性代数                 | M             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 概率论与数理统计             | H             |   |   |   |   |   |   |   |   |   |   |   |   |   |   | M |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 工程力学                 |               | M |   |   | H |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 高等数学                 | H             |   |   |   | L |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 大学物理                 |               | M |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 大学物理实验               |               | M |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 普通化学                 |               | M |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 普通化学实验               |               | M |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 电路电子技术               |               |   |   |   |   |   |   |   | H |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|        | 电路电子技术实验             |               |   |   |   |   |   |   |   | H |   |   |   |   |   | M |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

| 课程名称                       |            | 安全工程专业毕业要求指标点 |   |   |   |   |   |   |   |   |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |    |    |    |   |   |   |   |  |
|----------------------------|------------|---------------|---|---|---|---|---|---|---|---|----|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|---|----|----|----|---|---|---|---|--|
|                            |            | 1             | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 3 | 3  | 3 | 3 | 4 | 4 | 4 | 5  | 5 | 5 | 6 | 6 | 6 | 7 | 7 | 7 | 8 | 8 | 9 | 9  | 9  | 1  | 1 | 1 | 1 | 1 |  |
|                            |            | -             | - | - | - | - | - | - | - | - | -  | - | - | - | - | - | -  | - | - | - | - | - | - | - | - | - | - | - | -  | -  | 0  | 0 | 0 | 1 | 1 |  |
|                            |            | 1             | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2  | 3 | 4 | 1 | 2 | 3 | 1  | 2 | 3 | 1 | 2 | 3 | 1 | 2 | 3 | 1 | 2 | 1 | 2  | 3  | -  | - | - | - | - |  |
|                            | 工程热力学与传热学  |               |   | M |   |   | M |   |   |   |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |    |    | 1  | 1 | 1 | 1 | 1 |  |
|                            | 流体力学       |               |   | M |   | H |   |   |   |   |    |   |   | L |   |   |    |   |   |   |   |   |   |   |   |   |   |   |    |    |    |   |   |   |   |  |
|                            | 环境保护与生态安全  |               |   |   |   |   |   |   |   |   |    |   |   |   |   |   |    |   |   | H |   |   | M |   |   |   |   |   |    |    |    |   |   |   |   |  |
| 专业<br>课                    | 安全科学导论     |               |   |   |   |   |   |   |   |   |    |   |   |   |   |   |    |   |   |   |   |   | M |   |   |   |   |   |    | MM |    |   |   |   |   |  |
|                            | 安全法规       |               |   |   |   |   |   |   |   |   |    |   |   |   |   |   |    |   | M |   |   |   | H |   |   |   |   |   |    |    |    |   | M |   |   |  |
|                            | 安全管理       |               |   |   |   |   |   |   | H |   |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   | H |   |    |    | M  |   | M |   |   |  |
|                            | 安全人机工程学    |               |   |   |   |   | M |   |   |   |    |   | H |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |    |    | M  |   |   |   |   |  |
|                            | 安全系统工程     |               |   |   | M |   |   | M |   |   |    |   |   |   |   |   |    |   | M |   |   |   |   |   |   |   |   |   |    |    | M  |   |   |   |   |  |
|                            | 防火防爆安全技术   |               |   | M |   |   | M |   |   |   |    |   |   |   | M |   |    | H |   |   |   |   |   |   |   |   |   |   |    |    |    |   |   |   |   |  |
|                            | 工业通风       |               |   |   | M |   |   |   |   |   |    |   |   |   | M |   |    |   | M |   |   |   |   |   |   |   |   |   |    |    |    |   |   |   |   |  |
|                            | 安全检测与智能监测  |               |   | M |   |   |   |   |   |   |    |   |   |   | H | H |    |   |   |   |   |   |   |   |   |   |   |   |    |    |    |   |   |   |   |  |
|                            | 锅炉压力容器安全技术 |               |   |   |   | H |   |   |   |   |    |   |   |   | M |   |    |   |   |   | M |   |   |   |   |   |   |   |    |    |    |   |   |   |   |  |
|                            | 职业危害与防治    |               |   |   |   |   |   |   |   | M |    |   |   | H |   |   |    |   |   |   |   | H |   |   |   |   |   |   |    |    |    |   |   |   |   |  |
|                            | 工程项目管理     |               |   |   |   |   |   |   |   |   |    |   |   |   |   |   |    |   |   | M |   |   |   |   |   |   |   |   |    |    | MM |   |   |   |   |  |
|                            | 电气安全工程     |               |   |   | M |   | M |   |   |   |    |   |   | M |   |   |    |   |   |   |   |   |   |   |   |   |   |   |    |    |    |   |   |   |   |  |
|                            | 岩土力学       | M             |   |   | M | H |   |   |   |   |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |    |    |    |   |   |   |   |  |
|                            | 机械安全技术     |               |   |   |   |   |   |   |   |   | H  |   |   |   |   | M |    |   |   |   |   |   |   |   |   |   |   |   |    |    |    | M |   |   |   |  |
|                            | 建筑火灾安全工程   |               |   |   | M |   | M |   |   | M |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |    |    |    |   |   | M |   |  |
|                            | 建筑施工安全技术   |               |   |   | M |   | M |   |   | M |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |    |    |    |   | M |   |   |  |
| 智慧安全技术                     |            |               |   |   |   |   |   |   |   |   |    |   |   |   |   | M |    | H |   |   |   |   |   |   |   |   |   |   |    |    |    |   |   | H |   |  |
| 实<br>践<br>教<br>学<br>模<br>块 | 工程训练       |               |   |   |   |   | M |   |   |   |    |   |   |   |   |   |    |   |   | M |   |   |   |   | M |   |   |   |    |    |    |   |   |   |   |  |
|                            | 安全工程专业实验   |               |   |   |   |   | H |   |   |   |    |   |   |   |   |   |    |   |   | M |   |   |   |   | M |   |   |   |    |    |    |   |   |   |   |  |
|                            | 认识实习       |               |   |   |   |   |   |   |   |   |    |   |   |   |   |   |    |   |   | H |   |   |   |   |   |   |   |   | M  |    |    |   |   |   |   |  |
|                            | 生产实习       |               |   |   |   |   | M |   |   |   |    |   |   |   |   |   |    |   |   |   | H |   |   | M |   |   | M |   |    |    |    |   |   |   |   |  |
|                            | 毕业实习       |               |   |   |   |   |   |   |   |   | H  |   |   |   |   |   |    |   |   | M |   |   |   |   |   | M |   | H |    |    |    |   |   |   |   |  |
|                            | 毕业设计（论文）   |               |   |   |   |   |   |   |   |   |    | M |   |   | H |   |    |   |   |   |   |   |   |   |   |   |   |   | MM |    |    |   |   | M |   |  |
|                            | 机械设计基础课程设计 |               |   |   |   |   |   |   |   |   | MM |   |   |   |   |   | MM |   |   |   |   |   |   |   |   |   |   |   |    |    |    |   |   |   |   |  |
|                            | 工业通风课程设计   |               |   |   |   |   |   |   |   |   | H  | M |   |   |   |   |    |   |   |   | M |   |   |   |   |   |   |   |    |    |    |   |   |   |   |  |
|                            | 防火防爆课程设计   |               |   |   |   |   | H |   |   | M |    |   |   |   |   |   | M  |   |   |   |   |   |   |   |   |   |   |   |    |    |    |   |   |   |   |  |
|                            | 安全评价设计     |               |   |   |   |   |   |   |   |   |    |   |   |   |   |   |    |   |   | H |   |   |   |   |   |   |   |   | M  |    |    | M |   |   |   |  |

| 课程名称   |        | 安全工程专业毕业要求指标点 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|--------|--------|---------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
|        |        | 1             | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 5 | 5 | 5 | 6 | 6 | 6 | 7 | 7 | 7 | 8 | 8 | 9 | 9 | 9 |
|        |        | -             | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
|        |        | 1             | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 1 | 2 | 3 | 1 | 2 | 3 | 1 | 2 | 3 | 1 | 2 | 1 | 2 | 3 |
| 素质拓展模块 | 创新创业教育 |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | M |   |   |   |
|        | 第二课程   |               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

十二、课程修读进程表

XII.Course review Process Map



十三、教学环节设置及学分分布表

(四年制本科 2025级 安全工程专业)

XIII. Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质             | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                                                                               | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双<br>学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|----------------------------|----------------|
|                  |                      |                  |                                                                                                                                |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                            |                |
| 平台               | 公共<br>课程<br>平台<br>课程 | 必修               | 5105017<br>思想道德与法治<br>Ideology and morality and rule of law                                                                    | 3      | 48     | 42      | 0   | 0   | 6                | 1      |                            |                |
|                  |                      |                  | 5103001<br>中国近现代史纲要<br>An Outline of Modern and Contemporary History of China                                                  | 3      | 48     | 42      | 0   | 0   | 6                | 2      |                            |                |
|                  |                      |                  | 5102001<br>马克思主义基本原理<br>Fundamentals of Marxism                                                                                | 3      | 48     | 44      | 0   | 0   | 4                | 3      |                            |                |
|                  |                      |                  | 5101005<br>毛泽东思想和中国特色社会主义理论体系概论<br>Theoretical system of socialism with Chinese characteristics                                | 3      | 48     | 32      | 0   | 0   | 16               | 4      |                            |                |
|                  |                      |                  | 5101006<br>习近平新时代中国特色社会主义思想概论<br>An Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era | 3      | 48     | 32      | 0   | 0   | 16               | 4      |                            |                |
|                  |                      |                  | 1401010<br>大学英语（一）<br>College English (I)                                                                                      | 2      | 32     | 32      | 0   | 0   | 0                | 1      |                            |                |
|                  |                      |                  | 1401062<br>大学英语（二）<br>College English(II)                                                                                      | 2      | 32     | 32      | 0   | 0   | 0                | 2      |                            |                |
|                  |                      |                  | 1401011<br>大学英语（三）<br>College English (III)                                                                                    | 2      | 32     | 32      | 0   | 0   | 0                | 3      |                            |                |
|                  |                      |                  | 1401064<br>大学英语（四）<br>College English (IV)                                                                                     | 1      | 16     | 16      | 0   | 0   | 0                | 4      |                            |                |
|                  |                      |                  | 1501882<br>体育(一)<br>Physical Education (I)                                                                                     | 1      | 26     | 26      | 0   | 0   | 0                | 1      |                            |                |
|                  |                      |                  | 1501883<br>体育(二)<br>Physical Education (II)                                                                                    | 1      | 34     | 34      | 0   | 0   | 0                | 2      |                            |                |
|                  |                      |                  | 1501884<br>体育(三)<br>Physical Education (III)                                                                                   | 1      | 34     | 34      | 0   | 0   | 0                | 3      |                            |                |
|                  |                      |                  | 1501885<br>体育(四)<br>Physical Education (IV)                                                                                    | 1      | 34     | 34      | 0   | 0   | 0                | 4      |                            |                |
|                  |                      |                  | 5106001<br>形势与政策<br>World Affairs and State Policy                                                                             | 2      | 64     | 64      | 0   | 0   | 0                | 1-8    |                            | 分散进行           |
|                  |                      |                  | 1501024<br>体质健康测试<br>Physical Health Test                                                                                      | 0      | 16     | 0       | 0   | 0   | 16               | 1-8    |                            | 分散进行           |
|                  | 通识<br>教育<br>课程       | 必修               | 5107001<br>国家安全教育<br>National Security Education                                                                               | 1      | 16     | 8       | 0   | 0   | 8                | 2      |                            |                |

教学环节设置及学分分布表

(四年制本科 2025级 安全工程专业)

Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质       | 课<br>程<br>编<br>码                                                          | 课<br>程<br>名<br>称                                                                | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |                |                                                                           |                                                                                 |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 平台               | 公共课程<br>通识教育课程 | 1306018                                                                   | 人工智能导论<br>Introduction to Artificial Intelligence                               | 2      | 32     | 20      | 0   | 12  | 0                | 1      |                        |                |
|                  |                | 2501016                                                                   | 大学生劳动教育<br>College student labor education                                      | 1      | 32     | 2       | 0   | 0   | 30               | 1-4    |                        | 分散进行           |
|                  |                | 2502006                                                                   | 大学生心理健康教育<br>Mental Health Education                                            | 2      | 32     | 24      | 0   | 0   | 8                | 1      |                        |                |
|                  |                | 2503001                                                                   | 职业生涯规划与就业指导<br>Career Planning and Employment Guidance                          | 1      | 16     | 16      | 0   | 0   | 0                | 2      |                        |                |
|                  |                | 2504005                                                                   | 军事理论<br>Military Theory                                                         | 2      | 36     | 36      | 0   | 0   | 0                | 2      |                        |                |
|                  |                | 8001003                                                                   | 大学生创新创业基础<br>Foundation of Innovation and Entrepreneurship for college students | 1      | 16     | 16      | 0   | 0   | 0                | 2      |                        |                |
|                  |                | 医学类1学分<br>Medicine 1 Academic Credit                                      |                                                                                 |        |        |         |     |     |                  |        |                        |                |
|                  |                | 人文社科类1学分<br>Humanity and Social Science 1 Academic Credit                 |                                                                                 |        |        |         |     |     |                  |        |                        |                |
|                  |                | 经济管理类1学分<br>Economic and Management 1 Academic Credit                     |                                                                                 |        |        |         |     |     |                  |        |                        |                |
|                  |                | 美育教育类2学分<br>Art Education 2 Academic Credit                               |                                                                                 |        |        |         |     |     |                  |        |                        |                |
|                  |                | 思想政治及新时代素质教育类2学分<br>Ideological and Political Education 2 Academic Credit |                                                                                 |        |        |         |     |     |                  |        |                        |                |
|                  | 专业学科基础课程       | 0702603                                                                   | 高等数学B(一)<br>Advanced Mathematics B (I)                                          | 4      | 64     | 64      | 0   | 0   | 0                | 1      |                        |                |
|                  |                | 0702604                                                                   | 高等数学B(二)<br>Advanced Mathematics B (II)                                         | 5      | 80     | 80      | 0   | 0   | 0                | 2      |                        |                |
|                  |                | 0703605                                                                   | 大学物理B(一)<br>College Physics B (I)                                               | 2.5    | 40     | 40      | 0   | 0   | 0                | 2      |                        |                |
|                  |                | 0302001                                                                   | 工程制图<br>Engineering Drawing                                                     | 2      | 32     | 32      | 0   | 0   | 0                | 1      |                        |                |
|                  |                | 0702603                                                                   | 高等数学B(一)<br>Advanced Mathematics B(I)                                           | 4      | 64     | 64      | 0   | 0   | 0                | 1      |                        |                |
|                  |                | 2206006                                                                   | 普通化学<br>General Chemisty                                                        | 2      | 32     | 32      | 0   | 0   | 0                | 1      |                        |                |

教学环节设置及学分分布表

(四年制本科    2025级    安全工程专业)

Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程性质     | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                               | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------|------------------|--------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |          |                  |                                                                                |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 平台               | 专业学科基础课程 | 必修               | 2253017<br>普通化学实验<br>Experiments in General Chemisty                           | 1      | 16     | 0       | 16  | 0   | 0                | 1      |                        |                |
|                  |          |                  | 0701606<br>工程力学B<br>Engineering Mechanics B                                    | 3      | 48     | 42      | 6   | 0   | 0                | 2      |                        |                |
|                  |          |                  | 0702604<br>高等数学B(二)<br>Advanced Mathematics B(II)                              | 5      | 80     | 80      | 0   | 0   | 0                | 2      |                        |                |
|                  |          |                  | 0703605<br>大学物理B(一)<br>College Physics B(I)                                    | 2.5    | 40     | 40      | 0   | 0   | 0                | 2      |                        |                |
|                  |          |                  | 0401039<br>电路电子技术<br>Electric Circuit and Electronic Technologies              | 3      | 48     | 48      | 0   | 0   | 0                | 3      |                        |                |
|                  |          |                  | 0450005<br>电路电子技术实验<br>Electric Circuit and Electronic Technologies Experiment | 1      | 16     | 0       | 16  | 0   | 0                | 3      |                        |                |
|                  |          |                  | 0702104<br>线性代数B<br>Linear Algebra B                                           | 2      | 32     | 32      | 0   | 0   | 0                | 3      |                        |                |
|                  |          |                  | 0703606<br>大学物理B(二)<br>College Physics B(II)                                   | 2      | 32     | 32      | 0   | 0   | 0                | 3      |                        |                |
|                  |          |                  | 0703607<br>大学物理实验B<br>Experiments of College Physics B                         | 1      | 24     | 0       | 24  | 0   | 0                | 3      |                        |                |
|                  |          |                  | 0702304<br>概率论与数理统计B<br>Probability and Mathematical Statistics(B)             | 2.5    | 40     | 40      | 0   | 0   | 0                | 4      |                        |                |
|                  |          |                  | 0304017<br>机械设计基础<br>Fundamentals of Mechanical Design                         | 3.5    | 56     | 50      | 6   | 0   | 0                | 5      |                        |                |
|                  |          | 选修               | 1306012<br>数据库技术及应用<br>Database Technology and Applications                    | 1.5    | 24     | 24      | 0   | 0   | 0                | 2      |                        |                |
|                  |          |                  | 1306013<br>数据库技术及应用实验<br>Experiment of Database Technology and Applications    | 1      | 24     | 0       | 0   | 24  | 0                | 2      |                        |                |
|                  |          |                  | 0105168<br>环境保护与生态安全<br>Environmental Protection and Ecological Safety         | 1      | 16     | 16      | 0   | 0   | 0                | 4      |                        | 限选             |
|                  |          |                  | 0105176<br>流体力学<br>Fluid mechanics                                             | 3      | 48     | 44      | 4   | 0   | 0                | 4      |                        | 限选             |
|                  |          |                  | 0105064<br>工程热力学与传热学<br>Engineering Thermodynamics and Heat Transfer           | 3.5    | 56     | 56      | 0   | 0   | 0                | 5      |                        | 限选             |

教学环节设置及学分分布表

(四年制本科 2025级 安全工程专业)

Offered Course and Distribution of Academic Credits

| 课 程 类 型 |        |    | 课程性质    | 课 程 编 码                                                           | 课 程 名 称 | 学 分 | 合 计 | 课 内 学 时 |    |    | 实践学时 | 学 期 | 是否辅修/双学位课程   | 先 修 课 程/备 注 |
|---------|--------|----|---------|-------------------------------------------------------------------|---------|-----|-----|---------|----|----|------|-----|--------------|-------------|
|         |        |    |         |                                                                   |         |     |     | 讲课      | 实验 | 上机 |      |     |              |             |
| 模块      | 专业课程模块 | 必修 | 0105020 | 安全科学导论<br>Introduction of Safety Science                          | 1       | 16  | 16  | 0       | 0  | 0  | 3    | 是   |              |             |
|         |        |    | 0105001 | 安全管理<br>Safety Administration                                     | 2.5     | 40  | 36  | 4       | 0  | 0  | 5    | 是   |              |             |
|         |        |    | 0105006 | 电气安全工程<br>Electrical Safety Engineering                           | 2       | 32  | 32  | 0       | 0  | 0  | 5    | 是   |              |             |
|         |        |    | 0105120 | 安全人机工程<br>Human Engineering                                       | 2       | 32  | 28  | 4       | 0  | 0  | 5    | 是   |              |             |
|         |        |    | 0105163 | 安全系统工程<br>Safety System Engineering                               | 2       | 32  | 32  | 0       | 0  | 0  | 5    | 是   |              |             |
|         |        |    | 0105170 | 安全法规<br>Safety Laws/Regulations                                   | 1       | 16  | 16  | 0       | 0  | 0  | 5    | 是   |              |             |
|         |        |    | 0105013 | 防火防爆安全技术<br>Fire Prevention and Explosion Protection Technologies | 2.5     | 40  | 36  | 4       | 0  | 0  | 6    | 是   |              |             |
|         |        |    | 0105022 | 安全检测与智能监测<br>Safety Detection and Intelligent Monitoring          | 2       | 32  | 32  | 0       | 0  | 0  | 6    | 是   |              |             |
|         |        |    | 0105015 | 工业通风<br>Industrial Ventilation                                    | 2       | 32  | 32  | 0       | 0  | 0  | 6    | 是   |              |             |
|         |        |    | 0105161 | 职业危害与防治<br>Occupational Hazards and Protection                    | 2       | 32  | 32  | 0       | 0  | 0  | 6    | 是   |              |             |
|         |        |    | 0105164 | 锅炉压力容器安全技术<br>Boiler and Pressure Vessel Safety Technology        | 2       | 32  | 32  | 0       | 0  | 0  | 6    | 是   |              |             |
|         |        |    | 0105160 | 机械安全技术<br>Mechanical Safety Technology                            | 2       | 32  | 28  | 4       | 0  | 0  | 7    | 是   |              |             |
|         | 专业选修课程 | 选修 | 0105171 | 安全工程专业英语<br>Specialized English of Safety Engineering             | 2       | 32  | 32  | 0       | 0  | 0  | 3    |     |              |             |
|         |        |    | 0101217 | 岩土力学<br>Geomechanics                                              | 2       | 32  | 28  | 4       | 0  | 0  | 4    |     |              |             |
|         |        |    | 0105162 | 建筑火灾安全工程<br>Building Fire Safety Engineering                      | 2.5     | 40  | 36  | 4       | 0  | 0  | 5    | 是   | 专业特色课程，至少3选2 |             |

教学环节设置及学分分布表 (四年制本科 2025级 安全工程专业)

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|---------|----------------|----------------|------|---------|-----------------------------------------------------|-----|----------|---------|-----|-----|----------|-----|------------|--------------|
|         |                |                |      |         |                                                     |     |          | 讲 课     | 实 验 | 上 机 |          |     |            |              |
| 模块      | 专业<br>课程<br>模块 | 专业<br>选修<br>课程 | 选修   | 0101220 | 工程项目管理<br>Project Management                        | 2   | 32       | 32      | 0   | 0   | 0        | 6   |            |              |
|         |                |                |      | 0101240 | 三维数字建模基础<br>Basis of 3D Digital Modeling            | 2   | 32       | 20      | 0   | 12  | 0        | 6   |            |              |
|         |                |                |      | 0105026 | 建筑施工安全技术<br>Building Construction Safety Technology | 2   | 32       | 32      | 0   | 0   | 0        | 6   | 是          | 专业特色课程，至少3选2 |
|         |                |                |      | 0105132 | 化工安全技术<br>Chemical Safety Technology                | 2   | 32       | 32      | 0   | 0   | 0        | 6   |            |              |
|         |                |                |      | 0105167 | 防排烟工程<br>Smoke Control Engineering                  | 2   | 32       | 32      | 0   | 0   | 0        | 6   |            |              |
|         |                |                |      | 0101204 | 矿井通风与安全<br>Mine Ventilation and Safety              | 2   | 32       | 26      | 6   | 0   | 0        | 7   |            |              |
|         |                |                |      | 0105017 | 智慧安全技术<br>Intelligent Safety Technology             | 2   | 32       | 20      | 0   | 12  | 0        | 7   | 是          | 限选           |
|         |                |                |      | 0105021 | 阻燃技术<br>Flame Retardant Technique                   | 2   | 32       | 32      | 0   | 0   | 0        | 7   |            |              |
|         |                |                |      | 0105159 | 安全经济学<br>Safety Economics                           | 2   | 32       | 32      | 0   | 0   | 0        | 7   |            |              |
|         |                |                |      | 0105172 | 矿山与冶金安全技术<br>Mine and Metallurgy Safety Technology  | 2   | 32       | 32      | 0   | 0   | 0        | 7   | 是          | 专业特色课程，至少3选2 |
|         |                |                |      | 0702019 | 数学建模<br>Mathematical Modeling                       | 2   | 32       | 32      | 0   | 0   | 0        | 7   |            |              |
|         |                |                |      | 0925139 | 灾害应急与救援A<br>Disaster Emergency and Rescue A         | 2   | 32       | 32      | 0   | 0   | 0        | 7   |            |              |
|         |                |                |      | 2301184 | 交通运输安全技术<br>Transportation Security Technology      | 2   | 32       | 32      | 0   | 0   | 0        | 7   |            |              |
|         | 实践<br>教学<br>模块 | 专业<br>实践<br>课程 | 必修   | 2504006 | 军事训练<br>Military Training                           | 2   | 112      | 0       | 0   | 0   | 112      | 1   |            |              |
|         |                |                |      | 1701008 | 工程训练B<br>Engineering Training B                     | 1.5 | 3<br>(周) | 0       | 0   | 0   | 3<br>(周) | 3   |            |              |
|         |                |                |      | 0105057 | 认识实习<br>Introductory Practice Experience            | 2   | 2<br>(周) | 0       | 0   | 0   | 2<br>(周) | 5   |            |              |

## 教学环节设置及学分分布表

(四年制本科 2025级 安全工程专业)

## Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质 | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                               | 学<br>分 | 合<br>计    | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------|------------------|--------------------------------------------------------------------------------|--------|-----------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |          |                  |                                                                                |        |           | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 模块               | 实践教学模块   | 专业实践课程<br>必修     | 0304010 机械设计基础课程设计<br>Course Project in Basics of Mechanical Design            | 1      | 2<br>(周)  | 0       | 0   | 0   | 2<br>(周)         | 5      |                        |                |
|                  |          |                  | 0105014 安全工程专业实验<br>Safety Engineering Professional Experiment                 | 1      | 24        | 0       | 24  | 0   | 0                | 6      |                        |                |
|                  |          |                  | 0105090 工业通风课程设计<br>Course Project in Industrial Ventilation                   | 1      | 2<br>(周)  | 0       | 0   | 0   | 2<br>(周)         | 6      | 是                      |                |
|                  |          |                  | 0105175 安全评价设计<br>Project in Safety Assessment                                 | 1      | 2<br>(周)  | 0       | 0   | 0   | 2<br>(周)         | 6      | 是                      |                |
|                  |          |                  | 0105058 生产实习<br>Production Practice                                            | 3      | 3<br>(周)  | 0       | 0   | 0   | 3<br>(周)         | 7      |                        |                |
|                  |          |                  | 0105109 防火防爆课程设计<br>Course Project in Fire Prevention and Explosion Protection | 1      | 2<br>(周)  | 0       | 0   | 0   | 2<br>(周)         | 7      | 是                      |                |
|                  |          |                  | 0105097 毕业实习<br>Pre-graduation Internship                                      | 2      | 2<br>(周)  | 0       | 0   | 0   | 2<br>(周)         | 8      |                        |                |
|                  |          |                  | 0105098 毕业设计(论文)<br>Undergraduate Project(Thesis)                              | 8      | 14<br>(周) | 0       | 0   | 0   | 14<br>(周)        | 8      | 是                      |                |
|                  | 素质拓展模块   | 创新创业教育<br>必修     | 创新创业实践3学分<br>Innovation Practices 3 Academic Credits                           |        |           |         |     |     |                  |        |                        |                |
|                  |          | 第二课堂<br>必修       | 第二课堂3学分<br>Second Classroom 3 Academic Credits                                 |        |           |         |     |     |                  |        |                        |                |

\*创新创业教育和第二课堂主要安排在小学期进行